

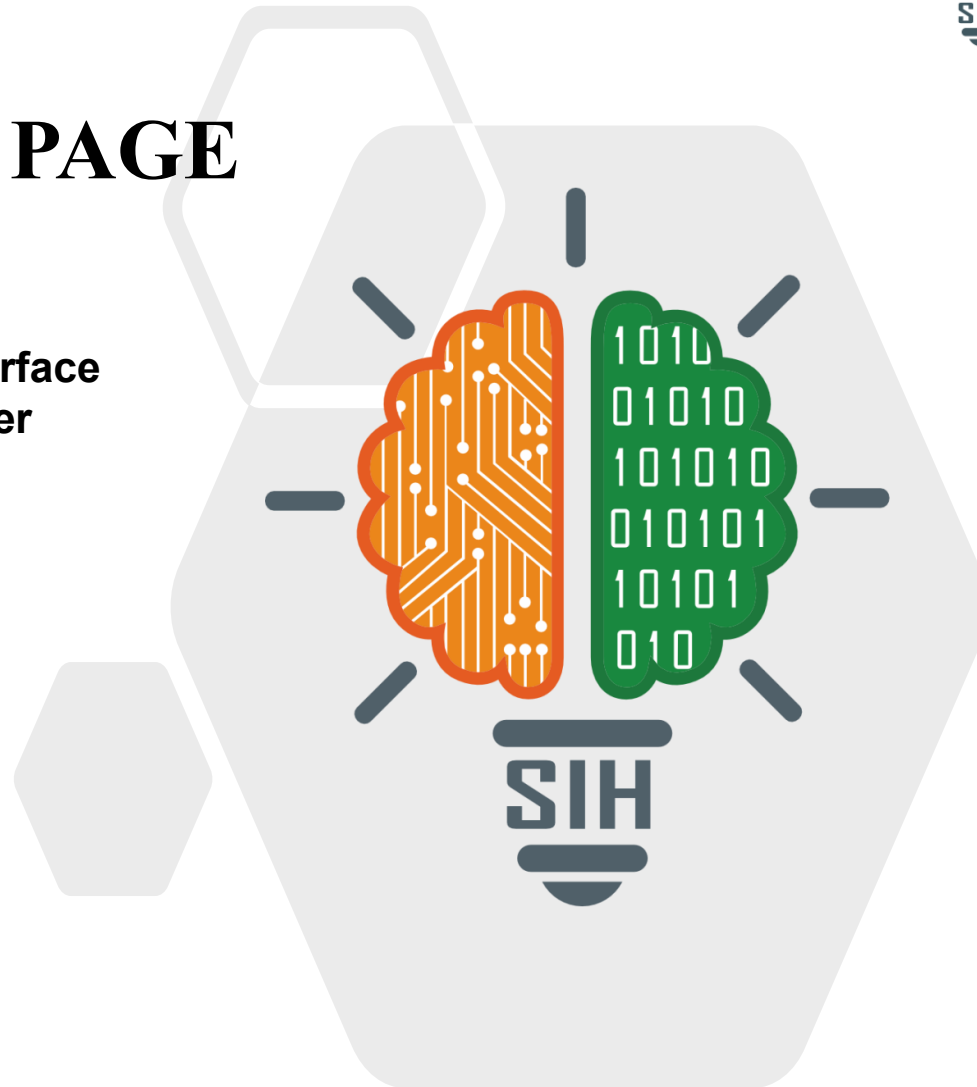
SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26120
- Problem Statement Title- Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation and Sucker Rod Pump Operations
- Theme- Smart Automation
- PS Category- Software
- Team ID- TBD
- Team Name (Registered on portal)- TEAM _____
- Organisation- Oil India Limited | Baghewala field, Rajasthan

“Physics + AI for India’s first cyclic-steam oilfield”



Your
Team
Name

Digital Twin for CSS and Sucker Rod Pump Optimization



Proposed Solution (Describe your Idea/Solution/Prototype)

For Oil India's Baghewala field — India's first cyclic-steam wells: a digital twin that recommends the steam, pump and stop settings for each cycle, and tells the engineer why.

Detailed explanation of the proposed solution



Every steam cycle, the twin re-plans the well from first principles — heat, thick oil, the rod pump.

One well, simulated day by day: inject → soak → produce

Tracks heat, viscosity, water cut and rod loads

Inputs: field parameters + Oil India's cycle records

Output: 5 settings + an operating rule, with a range

Field view: which of the 33 wells gets steam next

How it addresses the problem



Today's cycles are tuned by experience; the twin adds the physics and keeps the engineer in charge.

Manual steam and pump choices → one checked plan

Rods that float late in the cycle → slowed, then pulled

Steam wasted on cold oil → less steam per m³ of oil

No field data yet → every assumption tagged and ranged

Engineer reviews, confirms, then loads — advisory only

Innovation and uniqueness of the solution



Steam and pump treated as one system — because one thick oil causes both failures.

Couples the steam cycle to the rod pump — rare in tools

Rod float predicted from falling water cut, late in the cycle

Computed dynamometer card + a measured-card checker

Self-audited: 3 physics bugs found and fixed before judging

Offline, English + Hindi, no licence cost — built for the field

Links (demo video to be added)



Live dashboard:
baghewala-digital-twin.onrender.com



Demo video:
<https://> _____



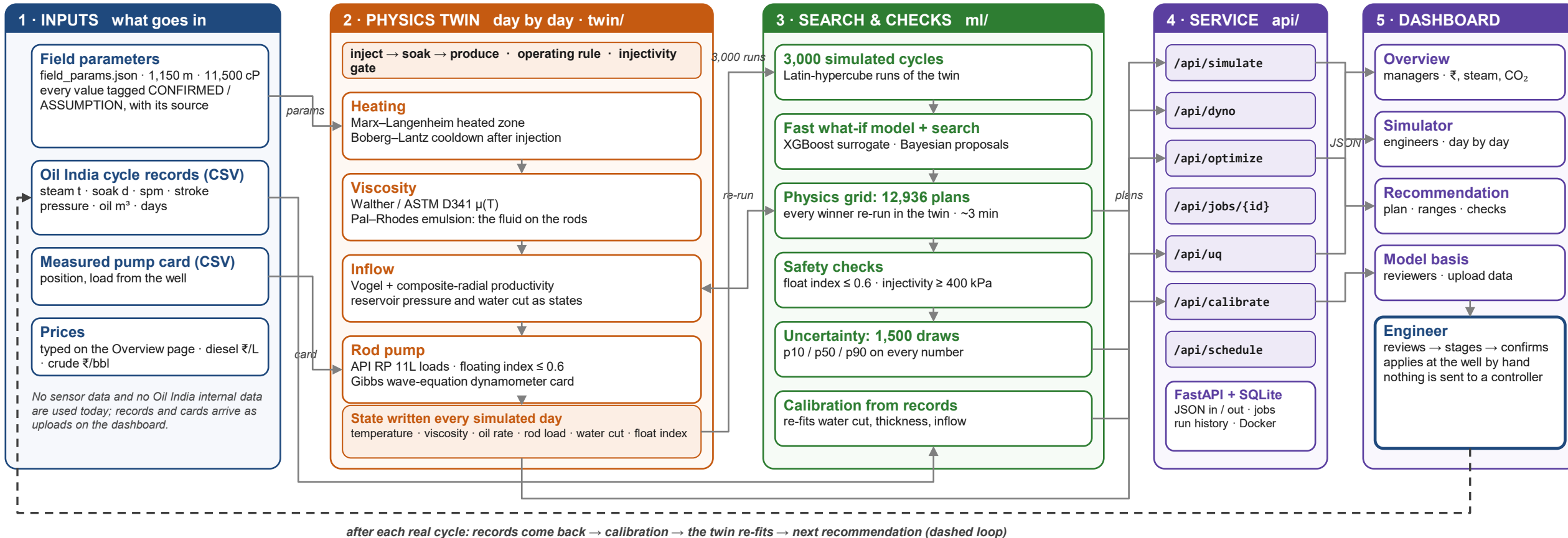
GitHub repository:
github.com/grv-io/sih2026-baghewala-digital-twin

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TECHNICAL APPROACH



- Technologies to be used: Python 3.13 · NumPy / SciPy · pandas · scikit-learn · XGBoost · scikit-optimize · FastAPI · Plotly · pytest
- Methodology and process for implementation: physics engine → 3,000 synthetic cycles → ML emulator → true-physics grid optimiser → dashboard → advisory set-points + operating rule to the operator (SCADA hook planned)



Built on published physics · benchmarked against a California steam-ratio band · 263 automated tests

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FEASIBILITY AND VIABILITY



Feasibility and viability — a working prototype, its known gaps, and how each gap closes with Oil India's data.

Analysis of the feasibility of the idea



Built and running today —
published physics,
real field numbers.

Real Baghewala inputs: 1,150 m, 11,500 cP

Steam 85–97 kgf/cm², 60–70 % quality (published)

Inside real field bands (uplift 5–6×, CalGEM)

Runs on a laptop · deployable API + Docker

Potential challenges and risks



Some numbers stay assumptions
until Oil India
shares its data.

No public per-cycle Baghewala records

Skin, water cut, condensate return unknown

Rod float depends on emulsion behaviour

Marginal at \$65/bbl; negative net of royalty + cess

Strategies for overcoming these challenges



Every assumption is tagged,
given a range,
and re-fitted from data.

Data request to Oil India ready (10 items)

Calibration loop: 10 cycles → re-fit in minutes

Uncertainty shown as a range, not one number

Advisory mode first · control hook later

Roadmap



Now: prototype,
263 tests, live demo



Oct: Oil India data,
calibrate the twin



Finale: field-checked
recommendations



Pilot: one well,
advisory mode

Team: Chemical Engineering + Software, MNIT Jaipur · open-source stack, no licence cost to Oil India

Impact and benefits — what changes for Oil India's operators, and what we honestly do and do not claim yet.

Potential impact on the target audience



The plan: 1,000 t steam · 89 kgf/cm²
64-in stroke · start 4.5 spm ·
slow at the float limit, then pull.

Steam per m³ of oil: 3.3 → 2.8 t (−14 %), same policy

Beats our assumed baseline in 84–96 % of runs

₹ gain ₹0–5k/day — set by the VFD's minimum speed

Drive slows the pump at the float limit, then pull

Field view: which well gets steam next

Benefits of the solution (social, economic, environmental, etc.)



Economic

Less steam per barrel:
3.3 → 2.8 t/m³ (−14 %)

23 % less steam · 11 % less oil

₹ gain hinges on OIL's VFD floor speed

A range is shown, never one number



Environmental

Every tonne of steam burns
71 kg of diesel

Fewer tonnes of steam per m³ oil

Less diesel and CO₂ per barrel

No wasted pumping after float



Social & operational

Operators keep control;
the twin explains itself

Plain words, English and Hindi

Rod failures not priced yet (\$15–50k)

Open source — skills stay with OIL

Assumptions to confirm with Oil India

Synthetic data today — the first real
cycle records re-fit the twin in minutes

Cash-positive in about half of scenarios
at OIL's FY25 price (vs a shut-in well)

Break-even at \$65/bbl; negative net of
royalty + cess — OIL's price deck decides

VFD minimum speed and pull criterion set
the ₹ gain — both on our data request

Scales with the field: 218 t (FY17) → 43,773 t (FY26) of heavy oil · 19 steam jobs in FY26 · one twin per well

Everything we used is public and cited — physics from the 1950s–80s, field facts from Oil India and SPE, benchmarks from California's regulator.

Details / Links of the reference and research work

Marx & Langenheim (1959). Reservoir heating by hot fluid injection. Trans. AIME 216. https://doi.org/10.2118/1266-G	SPE-23APOG-535203 (2023). Baghewala heavy oil: depth, API, viscosity, heater case. https://onepetro.org/SPEAPOG/proceedings-abstract/23APOG/2-23APOG/535203
Boberg & Lantz (1966). Thermally stimulated well rate — via SPE PEH ch. 15 (Prats). https://petrowiki.org/PEH:Thermal_Recovery_by_Steam_Injection	Yasin et al. (2022). Scientific Reports 12:11086 — Baghewala-1 reservoir properties. https://doi.org/10.1038/s41598-022-14831-5
Vogel (1968). Inflow performance relationships for solution-gas drive wells. JPT. https://onepetro.org/JPT/article/20/01/83/163252	Oil India Ltd — Rajasthan fields (BGW-8 CSS); investor presentation (realised price). https://www.oil-india.com/rajasthan-fields
Gibbs (1963). Predicting the behavior of sucker-rod pumping systems. JPT 15. https://doi.org/10.2118/588-PA	CalGEM (2021). Annual Report of the State Oil & Gas Supervisor — field steam-oil ratios. https://www.conservation.ca.gov/calgem/Documents/2021%20CalGEM%20Supervisor%20Annual%20Report.pdf
ASTM D341 viscosity–temperature charts (Walther); Pal & Rhodes (1989) emulsions. https://www.astm.org/Standards/D341.htm	Inside Climate News (2022) mirror of CalGEM injection data — 9,692 cyclic-steam records. https://github.com/InsideClimateNews/2022-09-ca-kern-oil-water
API Spec 11E / RP 11L — pumping-unit ratings and rod-string design. https://www.api.org/products-and-services/standards	Landscape: XSPOC, Lufkin SAM/SROD, ForeSite, AVEVA — none couples CSS with the pump. https://www.lufkin.com/solutions-services/srod/

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Every link is clickable in this PDF · full list with notes in the repository (docs/research/references.md)